



ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU, NEPAL

Phone: 01-5321365, 01-5344636

Email: ktm@sxc.edu.np

TITLE - SHEET

Name : Swikrit Wagle

Roll No : 948

Grade : 11

Section : I

Subject : Computer

S.N.	Unit/Chapter	Sub-Topics	☒	Signed	Date	Remarks
1	Computer System	1.1 Components	☐			
		1.2 Bus System	☐			
		1.3 Motherboard	☐			
		1.4 Hardware Interfaces	☐			
2	Number System and Boolean Logic	2.1 Number Conversion	☐			
		2.2 Boolean Logics	☐			
		2.3 Laws of Boolean Algebra	☐			
3	Operating System	3.1 System Configurations	☐			
		3.2 File Explorer Management	☐			
		3.3 Command Line Interface	☐			
4	Application Package	4.1 Word Processing Software	☒		20 th July, 2026 23 rd August, 2026	
		4.2 Spreadsheet Software	☒			
		4.3 Spreadsheet Formulas	☐			
		4.4 Presentation Software	☐			
5	Programming Concepts and Logics	5.1 Basic Syntax Framework	☐			
		5.2 C Tokens	☐			
		5.3 Operators & Expressions	☐			
		5.4 Control Structures	☐			
		5.5 Array 1D and 2D	☐			

		5.6 String Handling Functions				
		5.7 Mathematical Problems				
6	Web Technology I	6.1 Document Structure				
		6.2 Core Text elements				
		6.3 Lists & Hyperlinks				
		6.4 Table Structure				
		6.5 Form Elements				
		6.6 Inline & Internal Styling				
		6.7 External Style Sheets				
		6.8 Modern Layout Properties				
7	Multimedia	7.1 Multimedia in Web Page				
8	Information Security and Cyber Law	8.1 Information Security (CIA)				
		8.2 Cryptography				
		8.3 Security Measures				
		8.4 Cyber Crimes				
		8.5 Cyber Laws				

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LAB ASSIGNMENT 4.2

“SPREADSHEET SOFTWARE- MS EXCEL”

Submitted By	Submitted To	Signature
Name : Swikrit Wagle Roll No : 948 Class : 11 Section : I	Department of Computer Science, NEB (Plus Two) St. Xavier's College	

Submission Date: 23rd August, 2026

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Cell in Microsoft Excel

A cell is the basic unit of a worksheet in Microsoft Excel where data such as text, numbers, or formulas can be entered. It is formed by the intersection of a row and a column.

Example: The cell at the intersection of column B and row 5 is called B5.

Relative Cell Reference

A relative cell reference is a cell reference that changes automatically when a formula is copied or moved to another cell.

Example:

If cell C2 contains:

=A2+B2

and the formula is copied to C3, it automatically becomes:

=A3+B3

Here, A2 and B2 are relative references.

Mixed Cell Reference

A mixed cell reference is a cell reference in which either the row or the column is fixed, while the other part can change when the formula is copied. It is represented using the \$ symbol.

Example:

=\$A2

Here, column A is fixed, but the row can change.

If copied from B2 to B3, it becomes:

=\$A3

Another example is:

=A\$2

Here, row 2 is fixed, but the column can change.

Absolute Cell Reference

An absolute cell reference is a cell reference in which both the column and row remain fixed when a formula is copied to another cell. It is represented by using the \$ symbol before both the column and row.

Example:

=A\$2

If this formula is copied to another cell, the reference will still remain \$A\$2.

Example in calculation:

If cell B2 contains a tax rate and we want to use the same tax rate for many calculations:

=A3*\$B\$2

Here, \$B\$2 remains fixed when the formula is copied.

VLOOKUP

VLOOKUP (Vertical Lookup) is a Microsoft Excel function used to search for a value in the first column of a table and return a related value from another column in the same row.

For example, if a table contains:

Roll No.	Name	Marks
101	Ram	85
102	Sita	93
103	Hari	78

Table 1: VLOOKUP Example List

VLOOKUP can be used to find the marks of a student by searching for their Roll No.

Syntax of VLOOKUP

Syntax:

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

The arguments are:

Argument	Meaning
lookup_value	The value that Excel searches for in the first column of the table.
table_array	The range of cells containing the lookup table.
col_index_num	The column number from which the result should be returned.
range_lookup	Determines whether the match should be exact or approximate. FALSE means exact match and TRUE means approximate match.

Table 2: Meaning of Arguments

Example:

Suppose the student database is in A2:C11, where:

- Column A = Roll No.
- Column B = Student Name
- Column C = Marks

To find the marks of the student whose Roll No. is entered in E2:

=VLOOKUP(E2,A2:C11,3,FALSE)

Explanation:

- E2 → Roll number to search
- A2:C11 → Table containing student information
- 3 → Return the value from the 3rd column (Marks)
- FALSE → Find an exact match

So, if E2 = 105, Excel searches for Roll No. 105 in column A and returns the corresponding marks from column C.

Tabs in MS Excel

Formula Tab

The Formula tab contains tools for creating, managing, and checking formulas and functions. It includes options such as Insert Function, Function Library, Defined Names, Formula Auditing, and Calculation.

Data Tab

The Data tab provides tools for managing and analyzing data. It includes features such as Sort, Filter, Data Validation, Remove Duplicates, and What-If Analysis.

Practical Task

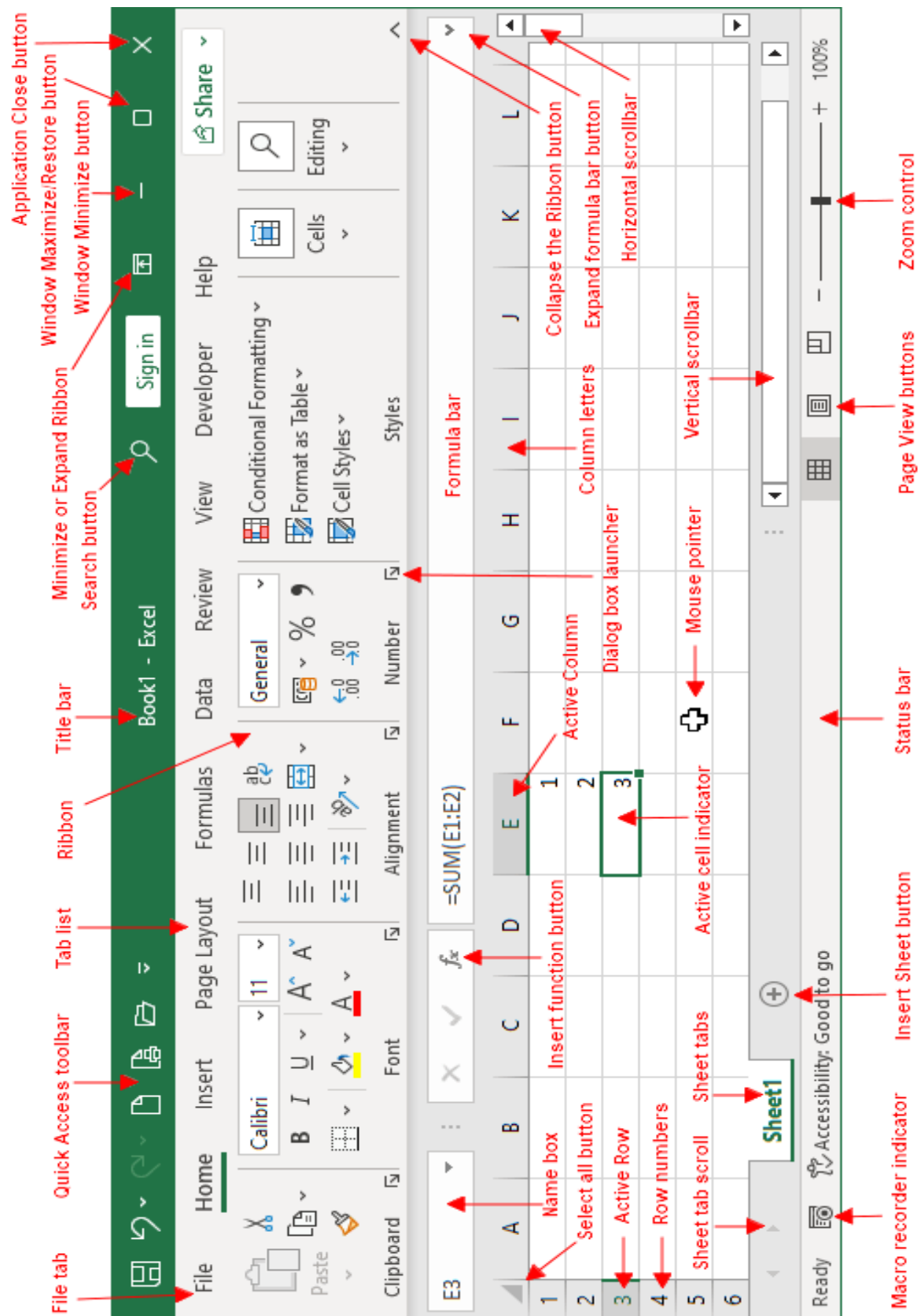


Figure 4.1: Structural Blueprint of the Microsoft Excel User Interface

ST. XAVIERS COLLEGE																				
Roll No.	Name	Section	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Maths Th	Maths Pr	Nepali Th	Nepali Pr	English Th	English Pr	Computer Th	Computer Pr	Total	Remark	Percentage	GPA	Grade	Rank
1	Sudhir Shah	I	75	25	74	23	73	22.5	68	22.5	68.5	23	47	42	563.5	Pass	93.92	3.76	A+	1
2	Ram Adhikari	G	64	22	46	23	68	22	67	20	63	22	42	4	463	Fail	77.17	NG	NG	NO RANK
3	Sanjana Tripathi	I	35	15	73	23	63	21	57	18	64	24	45	49	487	Pass	81.17	3.25	A	3
4	Daya Devkota	H	60	19	69	23	61	20	47	24	34	21	48	42	468	Pass	78	3.12	B+	6
5	Harichand Paudel	H	49	17	22	23	56	19	53	20	60	20	40	44	423	Fail	70.5	NG	NG	NO RANK
6	Devendra Raj Pandey	G	56	21	44	23	55	19.5	70	20	69	19	41	46	483.5	Pass	80.58	3.22	A	5
7	Hari Kishan Thakur	I	53	20	68	23	54	18	57	20	67	18	42	44	484	Pass	80.67	3.23	A	4
8	Ankit Gupta	I	68	19	38	7	62	20	60	19	62	22	45	32	454	Fail	75.67	NG	NG	NO RANK
9	Pranik Tandon	G	68	19	53	23	13	20	60	18	66	24	41	40	445	Fail	74.17	NG	NG	NO RANK
10	Saksham Regmi	H	59	19	55	23	66	20	62	22	63	27	41	43	500	Pass	83.33	3.33	A	2

Figure 4.2: Screenshot of Sheet 1: Student Mark sheet Calculation

First Terminal Examination Result						
Roll No.:						
Student's Name:	#N/A					
Section:	#N/A					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	#N/A	#N/A	#N/A	
Chemistry	75	25	#N/A	#N/A	#N/A	
Maths	75	25	#N/A	#N/A	#N/A	
Nepali	75	25	#N/A	#N/A	#N/A	
English	75	25	#N/A	#N/A	#N/A	
Computer	50	50	#N/A	#N/A	#N/A	
Total	#N/A					
Percentage	#N/A					
Grade	#N/A					
GPA	#N/A					
Remark	#N/A					
Rank	#N/A					

Figure 4.3: Screenshot of Sheet 2: VLOOKUP-Based Student Result

First Terminal Examination Result						
Roll No.:	1					
Student's Name:	Sudhir Shah					
Section:	I					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	75	25	4	
Chemistry	75	25	74	23	3.88	
Maths	75	25	73	22.5	3.82	
Nepali	75	25	68	22.5	3.62	
English	75	25	68.5	23	3.66	
Computer	50	50	47	42	3.56	
Total	563.5					
Percentage	93.92					
Grade	A+					
GPA	3.76					
Remark	Pass					
Rank	1					

Figure 4.4: Mark Sheet of Roll No.1

First Terminal Examination Result						
Roll No.:	2					
Student's Name:	Ram Adhikari					
Section:	G					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	64	22	3.44	
Chemistry	75	25	46	23	2.76	
Maths	75	25	68	22	3.6	
Nepali	75	25	67	20	3.48	
English	75	25	63	22	3.4	
Computer	50	50	42	4	1.84	
Total	463					
Percentage	77.17					
Grade	NG					
GPA	NG					
Remark	Fail					
Rank	NO RANK					

Figure 4.5: Mark Sheet of Roll No.2

First Terminal Examination Result						
Roll No.:	3					
Student's Name:	Sanjana Tripathi					
Section:	I					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	35	15	2	
Chemistry	75	25	73	23	3.84	
Maths	75	25	63	21	3.36	
Nepali	75	25	57	18	3	
English	75	25	64	24	3.52	
Computer	50	50	45	49	3.76	
Total	487					
Percentage	81.17					
Grade	A					
GPA	3.25					
Remark	Pass					
Rank	3					

Figure 4.6: Mark Sheet of Roll No.3

First Terminal Examination Result						
Roll No.:	4					
Student's Name:	Daya Devkota					
Section:	H					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	60	19	3.16	
Chemistry	75	25	69	23	3.68	
Maths	75	25	61	20	3.24	
Nepali	75	25	47	24	2.84	
English	75	25	34	21	2.2	
Computer	50	50	48	42	3.6	
Total	468					
Percentage	78					
Grade	B+					
GPA	3.12					
Remark	Pass					
Rank	6					

Figure 4.7: Mark Sheet of Roll No.4

First Terminal Examination Result						
Roll No.:	5					
Student's Name:	Harichand Paudel					
Section:	H					
SUBJECT	THEORY MARKS	PRACTICAL MARKS	OBTAINED MARKS IN TH	OBTAINED MARKS IN PR	GPA	
Physics	75	25	49	17	2.64	
Chemistry	75	25	22	23	1.8	
Maths	75	25	56	19	3	
Nepali	75	25	53	20	2.92	
English	75	25	60	20	3.2	
Computer	50	50	40	44	3.36	
Total	423					
Percentage	70.5					
Grade	NG					
GPA	NG					
Remark	Fail					
Rank	NO RANK					

Figure 4.8: Mark Sheet of Roll No.5

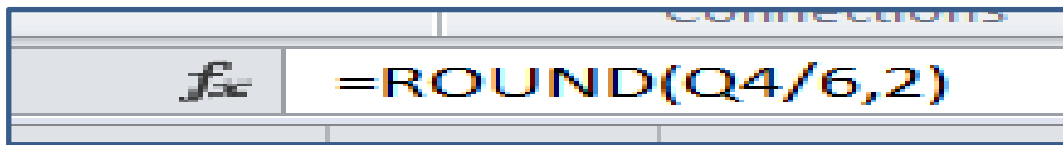


Figure 4.9: Screenshot of Percentage (Formula Bar)

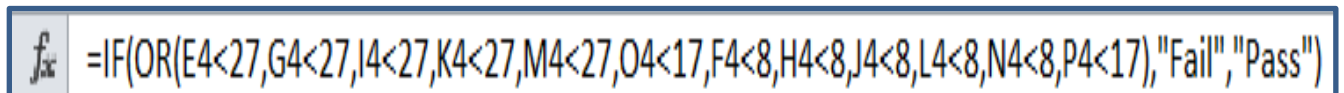


Figure 4.10: Screenshot of Remark (Formula Bar)

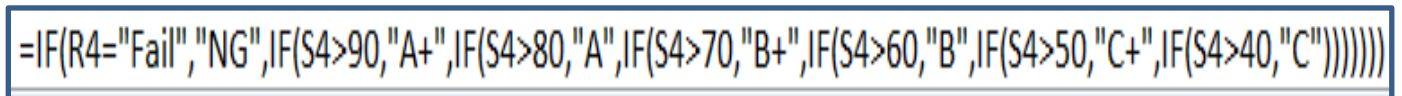


Figure 4.11: Screenshot of Grade (Formula Bar)

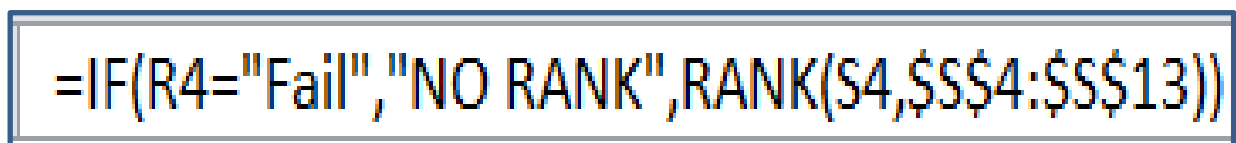


Figure 4.12: Screenshot of Rank (Formula Bar)

Lab Activity Reflection

- I learned how to use formulas and functions in MS Excel for calculations.
- I understood relative, absolute, and mixed cell references and their uses.
- I learned how to use VLOOKUP to search and retrieve information from a table.
- I learned to prepare a student marksheet using formulas and functions.
- I learned how to create a bill and perform automatic calculations such as total and amount.
- Overall, I improved my understanding of Excel for data calculation, organization, and presentation

Conclusion

This lab report helped me understand the basic concepts and practical uses of Microsoft Excel. I learned about the Excel interface, cell references, formulas, and the VLOOKUP function. I also gained practical experience in preparing a student marksheet, calculating marks and percentages, and connecting different sheets for easier data referencing. Overall, this practical improved my ability to use Excel for organizing data, performing calculations, and presenting information effectively.

